

Credit Card Approval Prediction

Phase 8: Project Demonstration

SmartBridge / SkillWallet Virtual Internship Program

Team Member	Role
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1. Demonstration Overview

This document walks through a live run of the Credit Card Approval Prediction application — from the empty applicant form, through a low-risk applicant profile (Approved), to a high-risk applicant profile (Rejected) — captured directly from the running Flask application.

2. Step 1 — Home Page / Applicant Form

Running `python app.py` and opening `http://127.0.0.1:5000` loads the applicant input form.

Credit Card Approval Prediction

AI-powered credit card approval prediction. Enter applicant details below.

Gender	Own Car
Female	Yes
Own Realty	Family Status
No	Married
Income Type	Education
Working	Higher Education
Housing Type	Annual Income
House / Apartment	e.g. 250000
Days Birth (age in days)	Days Employed
e.g. 12775	e.g. 1825
Family Members	Number of Loans
2	6
EMI Paid Off	EMI of Pastdues
4	0

Predict

Model: Random Forest (best of 4 models)

Figure 1 — Applicant input form (empty state).

3. Step 2 — Low-Risk Applicant → Approved

A pensioner applicant with stable income, higher education, no overdue EMIs, and a long, clean credit history is submitted. The model returns **Approved** with a high confidence score.

 **Credit Card Approval Prediction**

AI-powered credit card approval prediction. Enter applicant details below.

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Female	Yes
Own Realty	Family Status
No	Married
Income Type	Education
Working	Higher Education
Housing Type	Annual Income
House / Apartment	e.g. 250000
Days Birth (age in days)	Days Employed
e.g. 12775	e.g. 1825
Family Members	Number of Loans
2	6
EMI Paid Off	EMI of Pastdues
4	0

Predict

Approved — 78.06% confidence

Model: Random Forest (best of 4 models)

Figure 2 — Prediction result for a low-risk applicant.

4. Step 3 — High-Risk Applicant → Rejected

The same applicant profile is resubmitted with EMI_PASTDUES raised from 0 to 7 (multiple overdue payment months). The model correctly flips its decision to **Rejected**, demonstrating that the credit-history features are actually driving the prediction rather than being ignored.

 **Credit Card Approval Prediction**

AI-powered credit card approval prediction. Enter applicant details below.

Gender

Female

Own Car

Yes

Own Realty

No

Family Status

Married

Income Type

Working

Education

Higher Education

Housing Type

House / Apartment

Annual Income

e.g. 250000

Days Birth (age in days)

e.g. 12775

Days Employed

e.g. 1825

Family Members

2

Number of Loans

6

EMI Paid Off

4

EMI of Pastdues

0

Predict

Rejected — 68.66% confidence

Model: Random Forest (best of 4 models)

Figure 3 — Prediction result for a high-risk applicant.

5. Demonstration Checklist

Step	Verified
Dataset loads without errors (application_record.csv, credit_record.csv)	Yes
Preprocessing produces final_processed_dataset.csv with a realistic Approved/Rejected split	Yes
All 4 models train and print evaluation metrics	Yes
Best model (Random Forest) is saved to Flask/model/model.pkl	Yes
Flask app starts without errors and serves the home page	Yes
Low-risk applicant profile returns Approved	Yes
High-risk applicant profile returns Rejected	Yes
Confidence percentage displays correctly for both outcomes	Yes

6. How to Reproduce This Demo

- `pip install -r requirements.txt`
- Open and run all cells in Training/Credit_Card_Approval_Model_Training.ipynb (saves model.pkl, model_columns.pkl, scaler.pkl, encoders.pkl into Flask/model/).
- `cd Flask && python app.py`
- Open `http://127.0.0.1:5000` and submit the form with any applicant profile.